

TỔNG QUAN VỀ LẬP TRÌNH WEB

HUY INIT

Posts and Telecommunications Institute of Technology (PTIT)
Department of Computer Science





**Nguyen Quang
Huy, M.Sc.**

Academic Background

-  Member, Faculty of Information Technology, PTIT Mar. 2024 – Present
-  M.Sc. in Computer Science, PTIT May 2026
-  M.Sc. in Information Science, JAIST Sep. 2026

Research Interests

-  Computer Vision
-  Explainable Artificial Intelligence (XAI)
-  Recommender Systems
-  Continual Learning
-  Educational Technology (EdTech)

Contact Information

- **Github:** <https://github.com/huyinit>
- **Facebook:** <https://www.facebook.com/huyinit13>
- **Instagram:** <https://www.instagram.com/huyinit/>
- **Youtube Bài Giảng:** <https://www.youtube.com/@huyinit>
- **Youtube Vlog:** <https://www.youtube.com/@huyinitvlog>

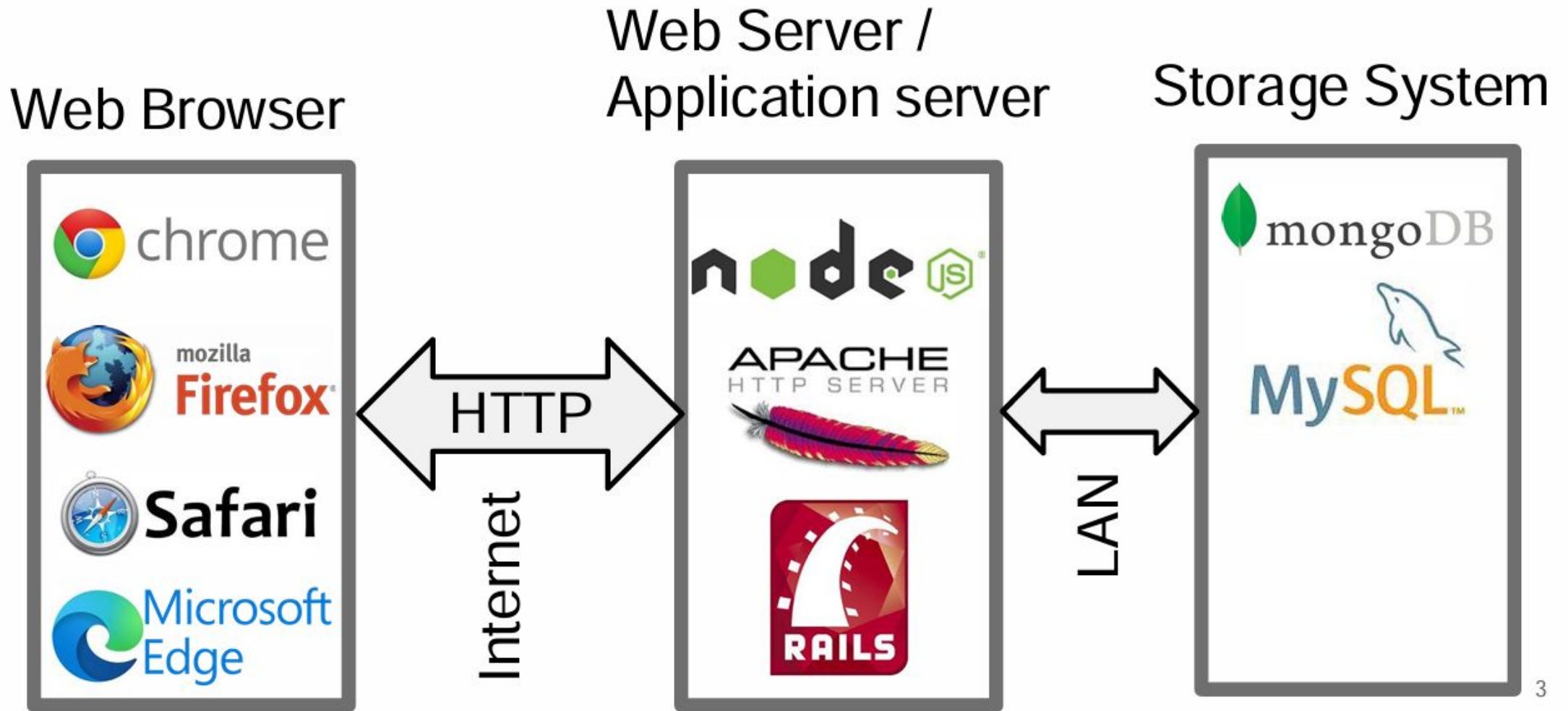
Seeking Research Collaboration

I welcome collaboration with researchers, graduate and undergraduate students, and industry professionals working in AI, RS, XAI, Continual Learning, and related fields. Whether you are interested in joint publications, undergraduate or graduate thesis projects, research internships, research discussions, or interdisciplinary collaborations, I would be happy to connect and explore new ideas together. Feel free to contact me at huyng@ptit.edu.vn.

Môn học: Phát triển Ứng dụng Web

- Giới thiệu các công nghệ và nền tảng để xây dựng ứng dụng web hiện đại.
- Tìm hiểu kiến trúc **Full Stack** : **Trình duyệt (Browser)** $\Leftrightarrow$ **Máy chủ Web (Web Server)** $\Leftrightarrow$ **Cơ sở dữ liệu (Database/Storage System)** .
- **Mục tiêu học tập:** Hiểu quy trình xây dựng, triển khai và vận hành một ứng dụng web hoàn chỉnh.
- Học thông qua thực hành (**Learning by Doing**) bằng cách phát triển các dự án thực tế.
- Làm quen với các công nghệ phổ biến:
 - **Frontend:** React.js
 - **Backend:** Node.js + Express.js, **Django (Python)**
 - **Database:** MongoDB (và giới thiệu tích hợp với các hệ quản trị CSDL khác)
- **Kết quả đạt được:** Có khả năng thiết kế, xây dựng và triển khai một ứng dụng web Full Stack, đồng thời hiểu cách các thành phần Frontend, Backend và Database phối hợp với nhau trong một hệ thống thực tế.

Full Stack Web Application Architecture

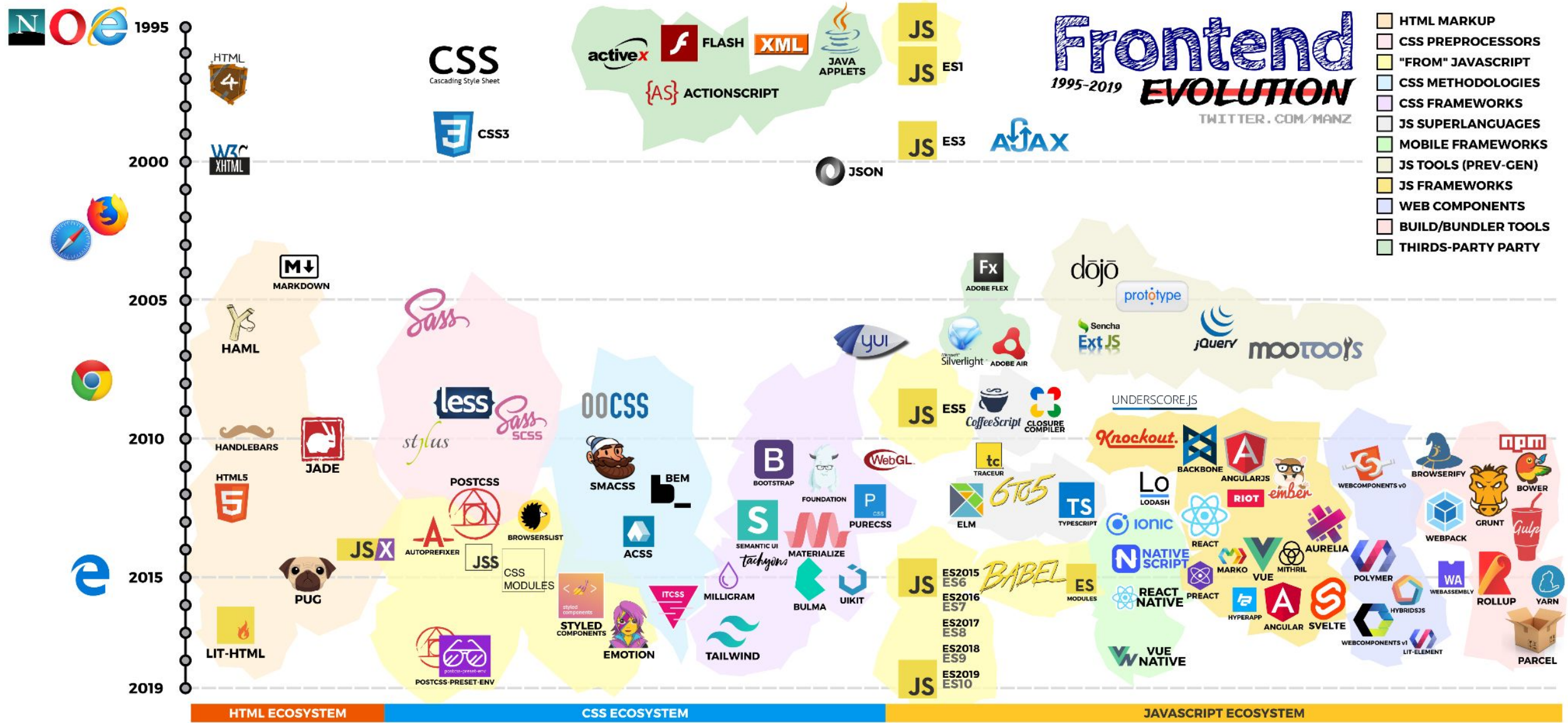


1. Frontend Development
2. Backend Development
3. Communication & APIs
4. Database Systems
5. DevOps & Cloud Computing
6. Web Security
7. Emerging Trends
& AI-Powered Web Development

1. *Front-end web development, also known as client-side development is the practice of producing some markup languages and other tools for a website or Web Application so that a user can see and interact with them directly.*
2. *Back-end development—also known as server-side development—is the architecture, code, and databases that power a website or web application behind the scenes. It focuses on the parts of the application that users don't see or interact with directly.*
3. *An API, or application programming interface, is a set of rules or protocols that enables software applications to communicate with each other to exchange data, features and functionality.*
4. *A database is an organized collection of structured information, or data, typically stored electronically in a computer system. A database is usually controlled by a database management system (DBMS).*
5. *DevOps is a modern software development approach that combines development (Dev) and operations (Ops) to streamline collaboration, accelerate delivery, and improve software quality.*

Cloud computing is the on-demand delivery of IT resources over the Internet with pay-as-you-go pricing. Instead of buying, owning, and maintaining physical data centers and servers, you can access technology services, such as computing power, storage, and databases, on an as-needed basis from a cloud provider.

1. *Web Application Security, often abbreviated as Web AppSec, is a set of practices, technologies, and processes designed to safeguard web applications from various threats, vulnerabilities, and attacks.*



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TOP 10 FRONTEND WEB DEVELOPMENT TOOLS IN 2026



React.js



Next.js



Vue.js



Angular



Svelte



Tailwind CSS



Astro



WebAssembly
(Wasm)



Vite



Figma

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- 2. **Backend Development**
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Back-end development—also known as server-side development—is the architecture, code, and databases that power a website or web application behind the scenes. It focuses on the parts of the application that users don’t see or interact with directly.

Framework	Language	Performance	Scalability	Community	Ease of Use
 django	 python	★★★★☆	★★★★☆	★★★★★	★★★★☆
 express JS	 node JS	★★★★★	★★★★☆	★★★★☆	★★★★☆
 spring boot	 Java	★★★★☆	★★★★★	★★★★☆	★★★★☆
 Laravel	 php	★★★★☆	★★★★☆	★★★★☆	★★★★☆
 RAILS	 Ruby	★★★★☆	★★★★☆	★★★★☆	★★★★☆
 ASP.NET Core	 C#	★★★★★	★★★★☆	★★★★☆	★★★★☆
 nest	 TS TypeScript	★★★★☆	★★★★★	★★★★☆	★★★★☆
 Flask	 python	★★★★☆	★★★★☆	★★★★☆	★★★★★
 koa	 node JS	★★★★★	★★★★☆	★★★★☆	★★★★☆
 Phoenix	 elixir	★★★★★	★★★★★	★★★★☆	★★★★☆

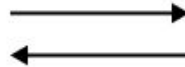
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End User with Browser



Request



Response



Server Back-end
System

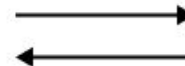


An API, or application programming interface, is a set of rules or protocols that enables software applications to communicate with each other to exchange data, features and functionality.



Customer

Make the order

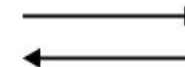


Delivery of
Order



Waiter

Take the Order

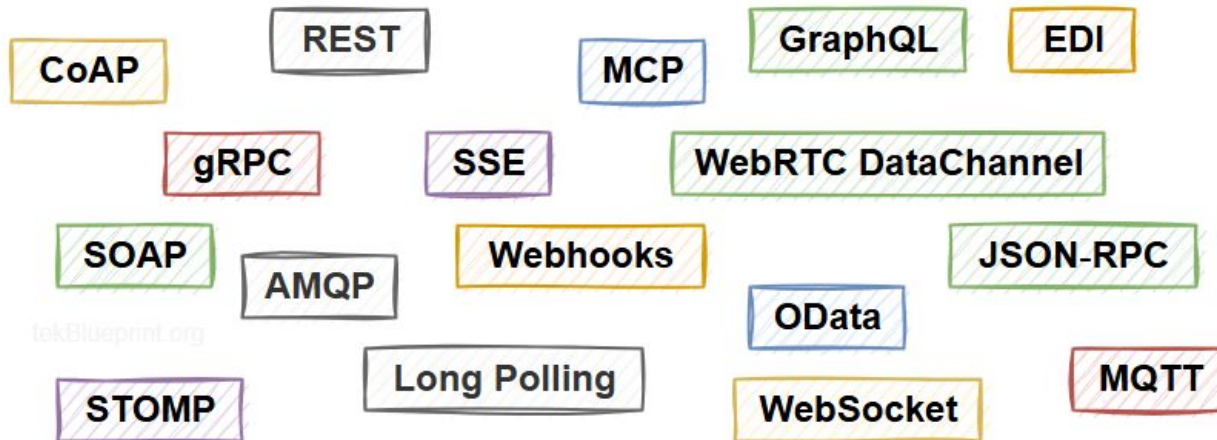


Bringing from
Kitchen



Chef

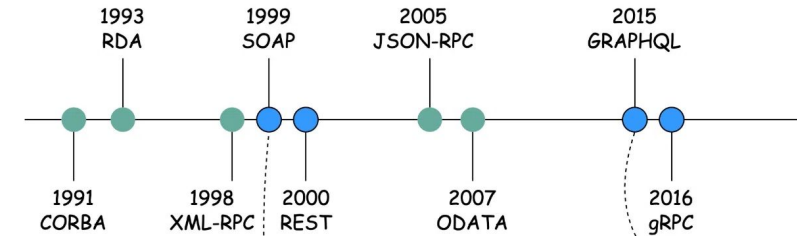
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[Technology] What happened when I looked into APIs | 単純な知的労働業務をなくす専門家 | Apient

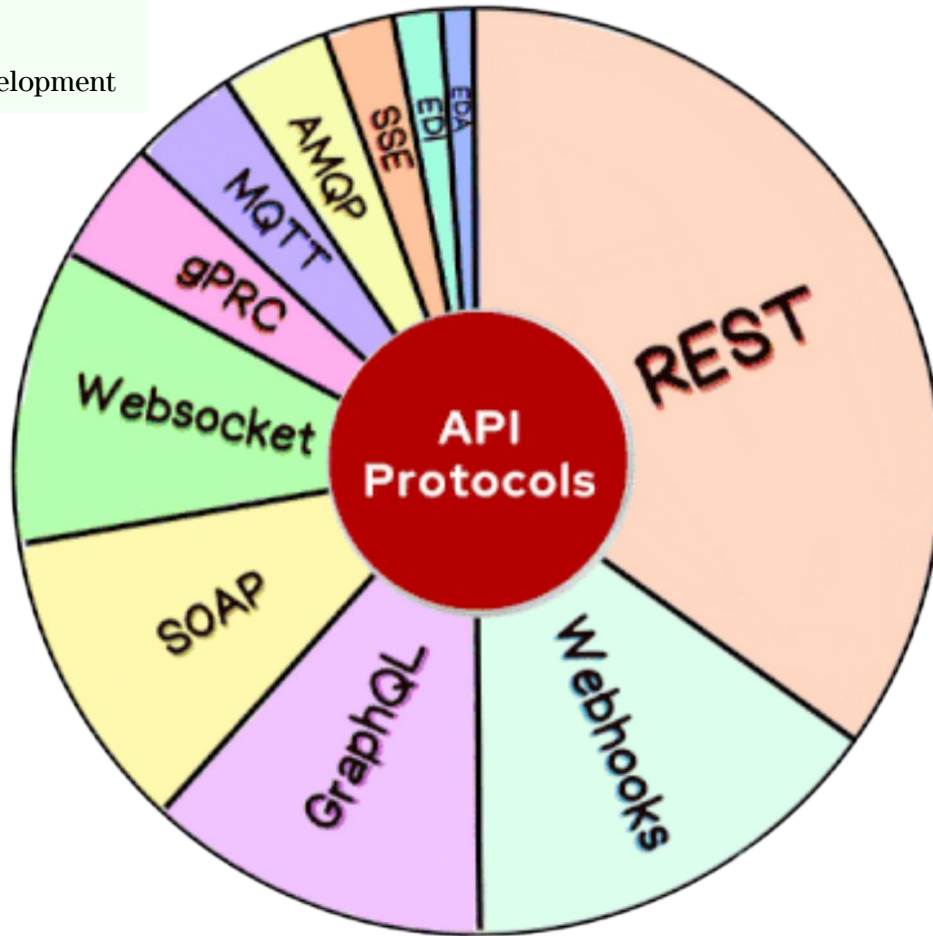
API Architectural Styles Comparison

Source: altexsoft



	SOAP (Simple Object Access Protocol)	REST (REpresentational State Transfer)	GraphQL	RPC (Remote Procedure Call)
Organized in terms of	enveloped message structure	compliance with six architectural constraints	schema & type system	local procedure call
Format	XML only	XML, JSON, HTML, plain text	JSON	JSON, XML, Protobuf, Thrift, FlatBuffers
Learning curve	Difficult	Easy	Medium	Easy
Community	Small	Large	Growing	Large
Use cases	- payment gateways - identity management - CRM solutions - financial and telecommunication services - legacy system support	- public APIs - simple resource-driven apps	- mobile APIs - complex systems - micro-services	- command and action-oriented APIs - high performance communication in massive micro-services systems

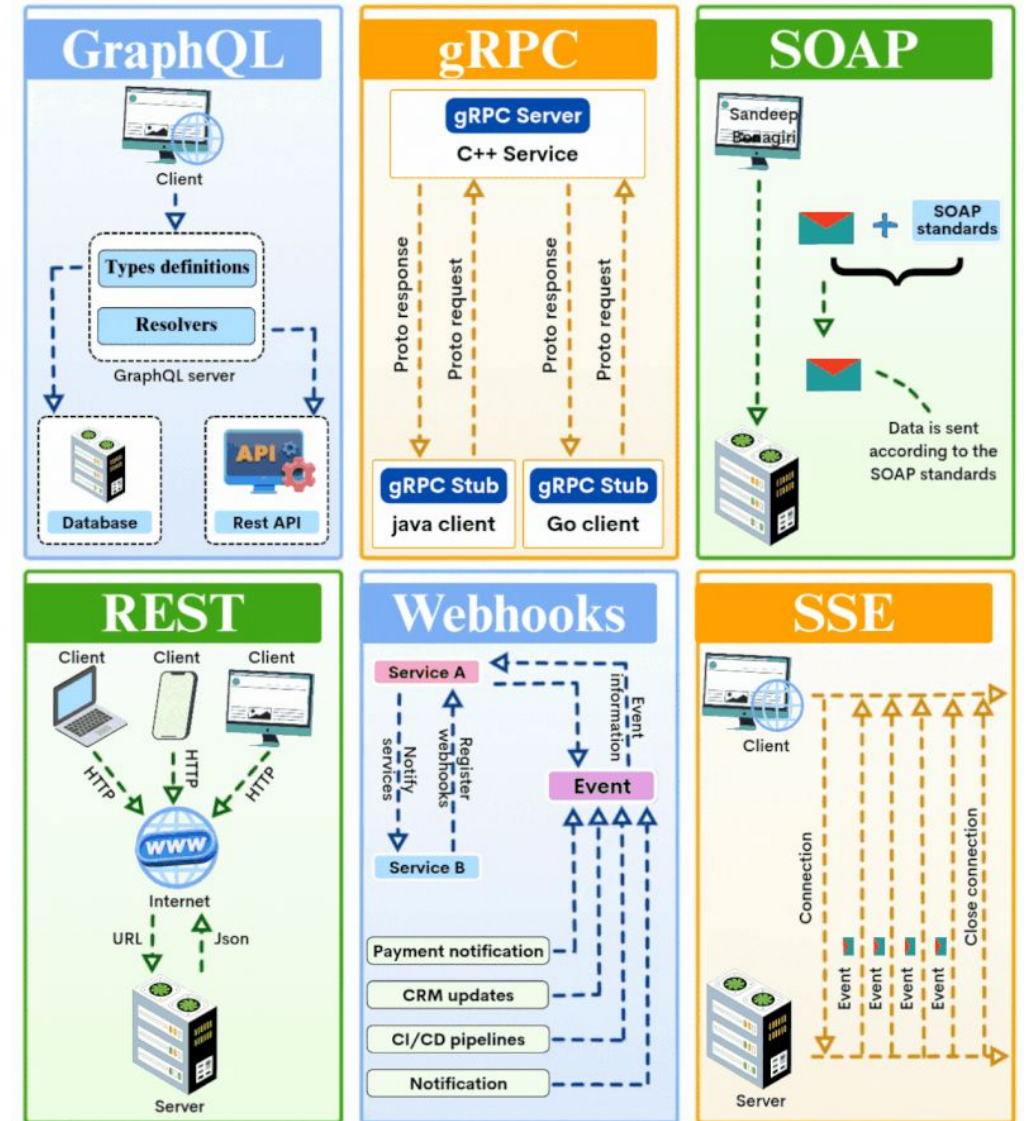
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API Protocols



Sandeep Bonagiri
Tech Career Coach



Helping you escape the 4–8 LPA trap | Link in comment section

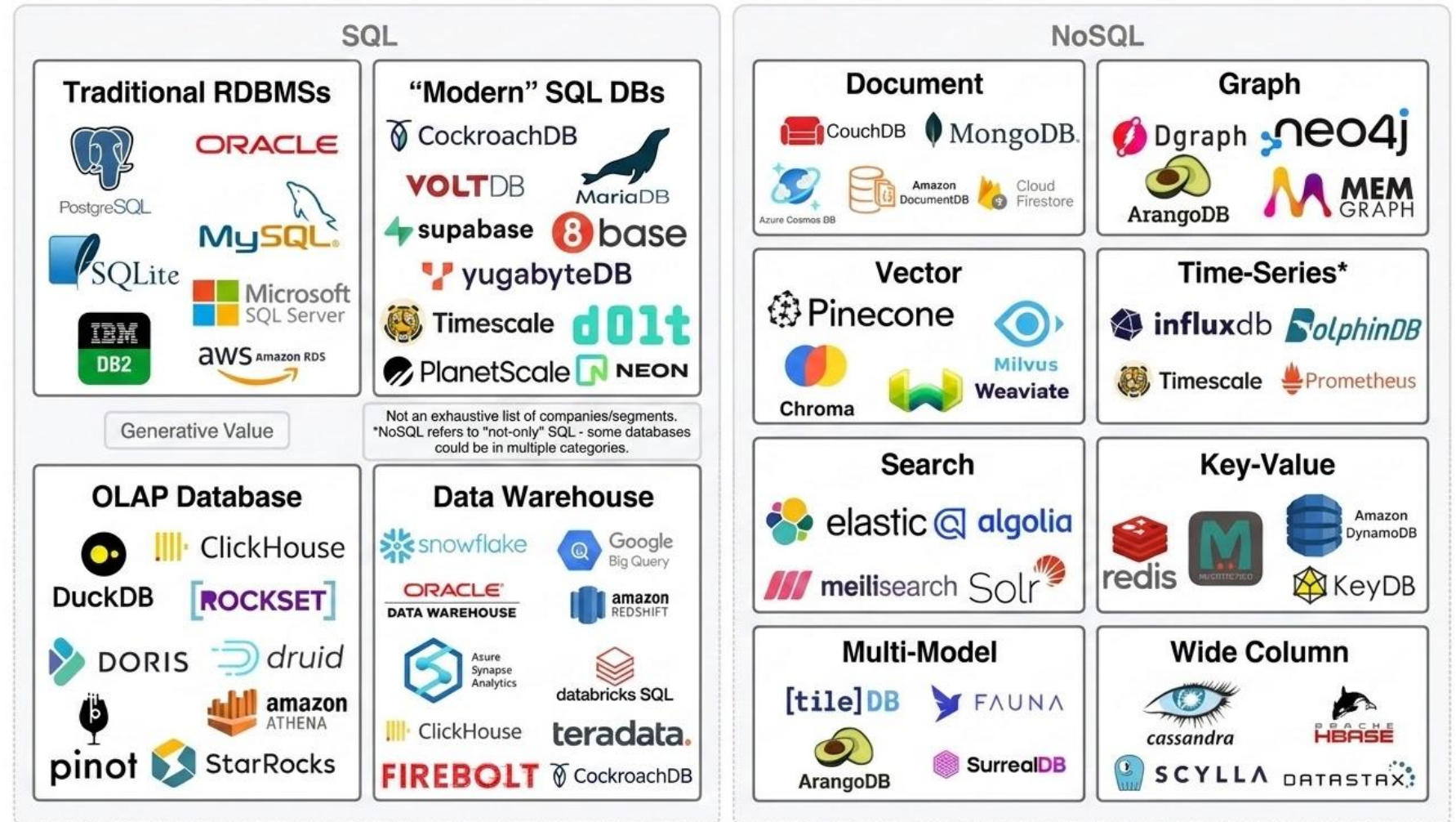


How APIs work and their types: A guide by Oliver | freeCodeCamp posted on the topic | LinkedIn

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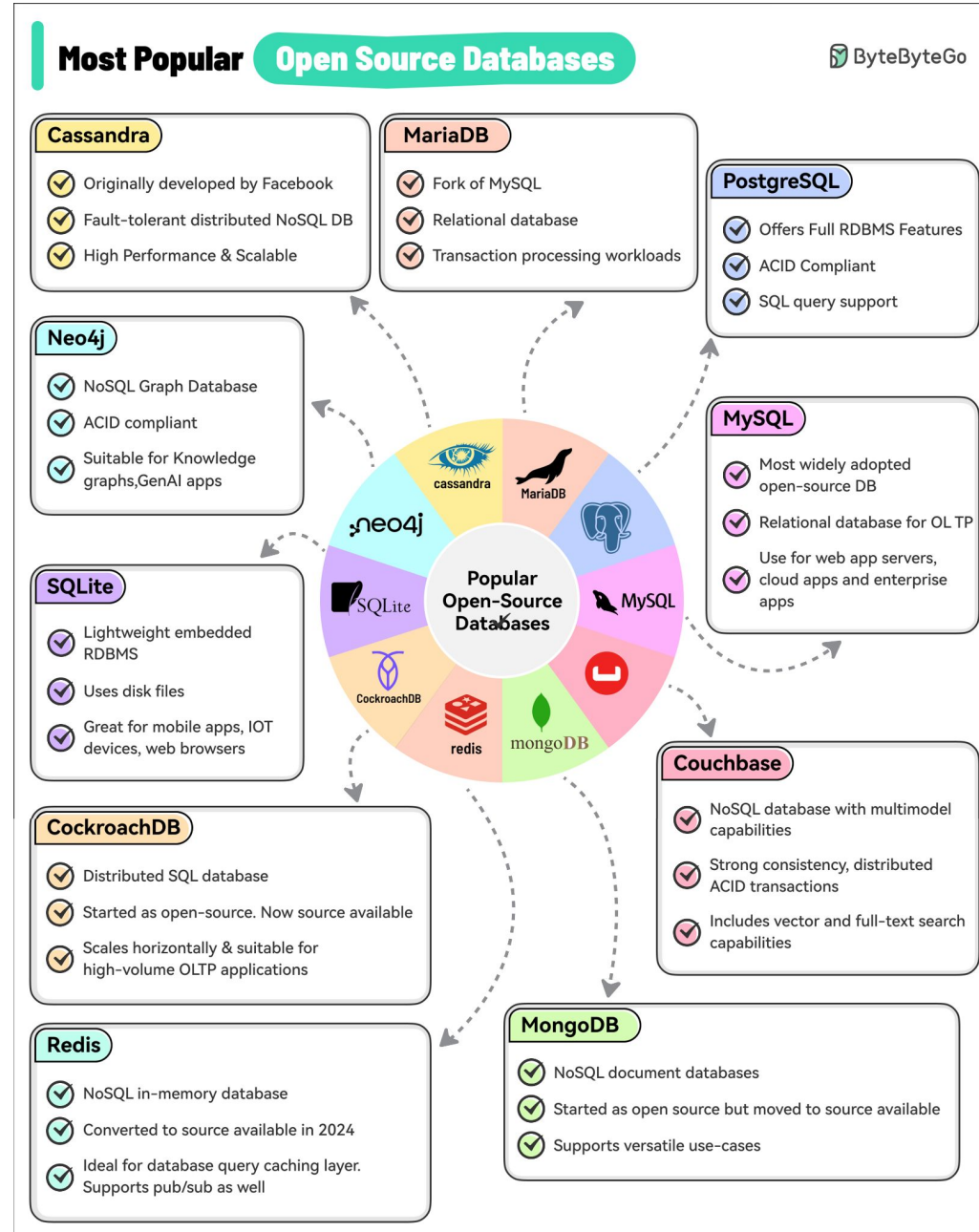
A database is an organized collection of structured information, or data, typically stored electronically in a computer system. A database is usually controlled by a database management system (DBMS).

Database Ecosystem



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"A cloud database is a database that typically runs on a cloud computing platform, and access to the database is provided as a service."

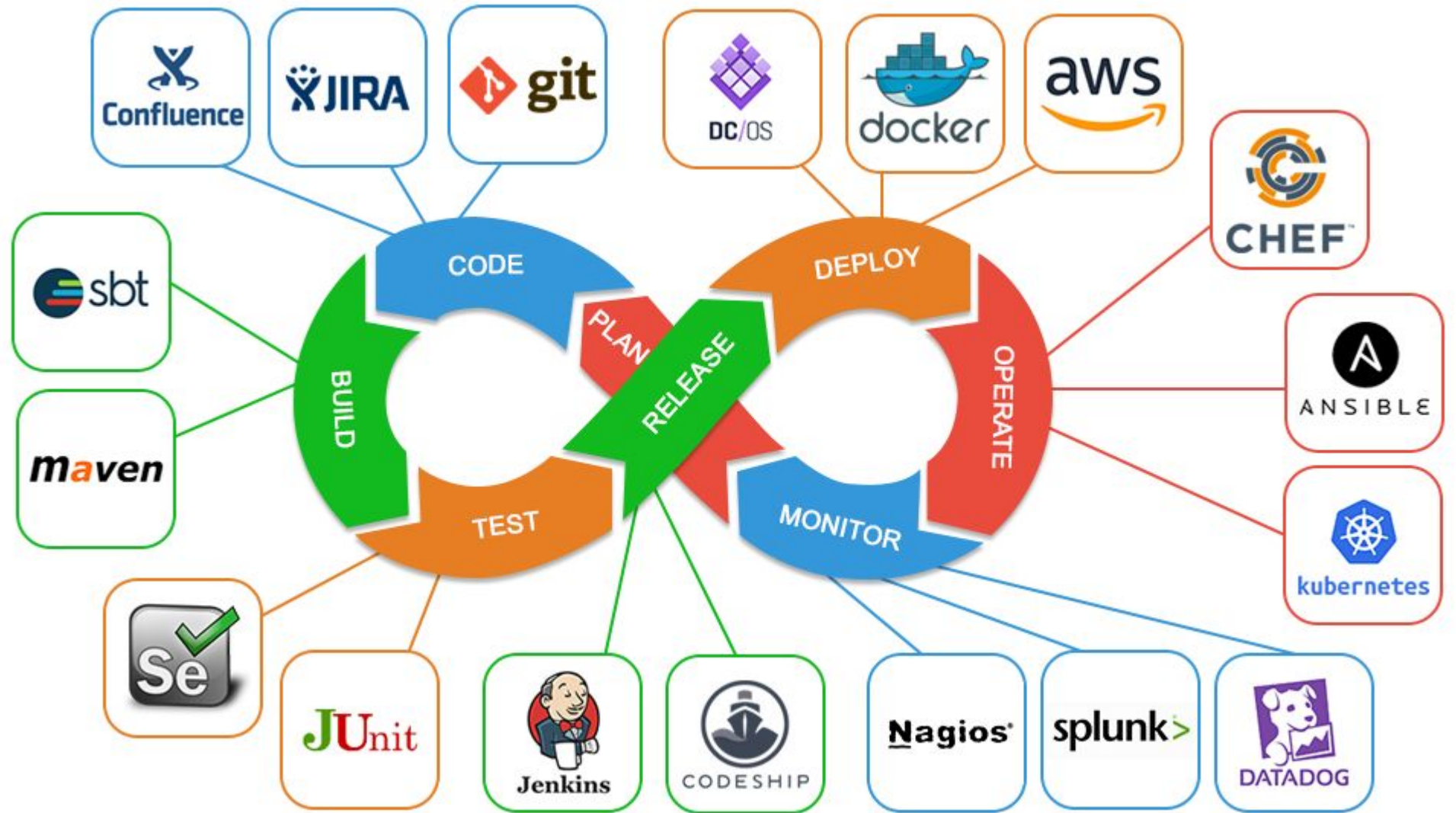
Cloud Database Cheat Sheet

 blog.bytebytego.com

DB Type		aws		Azure		Google Cloud		Open Source / 3rd Party	
Structured	Relational 	 RDS	 Redshift	 SQL Database	 Synapse Analytics	 Cloud SQL	 BigQuery	 Oracle	 PostgreSQL
	Columnar 							 MySQL	 SQL Server
Semi Structured	Key Value 	 DynamoDB		 Cosmos DB		 BigTable		 Redis	 Scylla
	In-Memory 	 ElastiCache		 Azure Cache for Redis		 Memory Store		 Redis	 Memcached
	Wide Column 	 Keyspaces		 Cosmos DB		 BigTable		 Cassandra	 Scylla
	Time Series 	 Timestream		 Time Series Insights		 BigTable		 Influx	 OpenTSDB
	Immutable Ledger 	 Quantum Ledger DB		 Confidential Ledger		 BigQuery		 Hyper Ledger Fabric	
	Geospatial 	 Keyspaces		 Cosmos DB		 BigTable		 PostGIS	 geomesa
	Graph 	 Neptune		 Cosmos DB		 CloudSpanner		 OrientDB	 Dgraph
	Document 	 Document DB		 Cosmos DB		 FireStore		 MongoDB	 Couchbase
	Text Search 	 OpenSearch		 Cognitive Search		 CloudSearch		 Elastic search	 Elassandra
	UnStructured	Blob 	 S3		 Blob Storage		 Cloud Storage		 Ceph

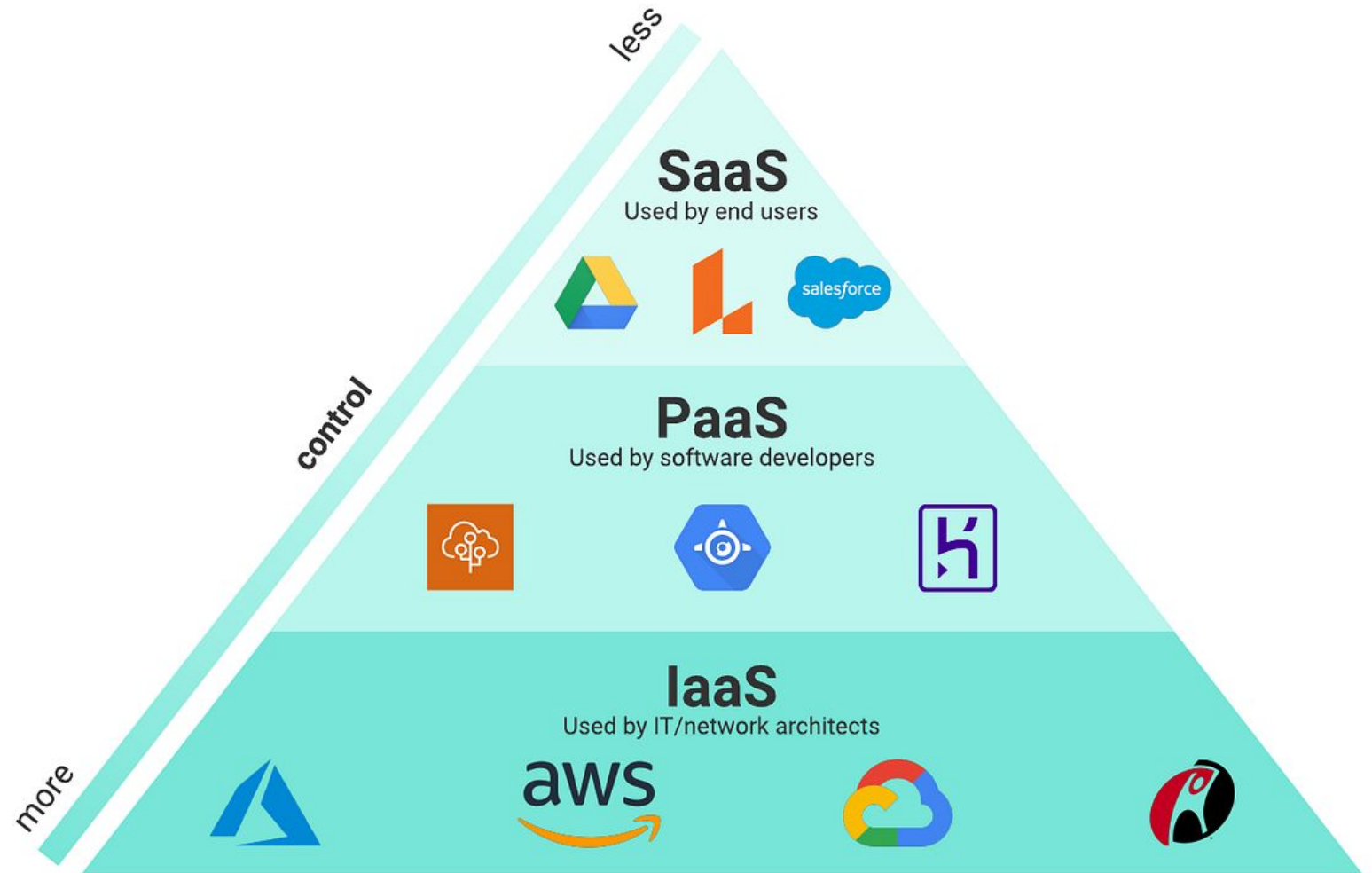
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Modification of Message



Denial of Services



Phishing



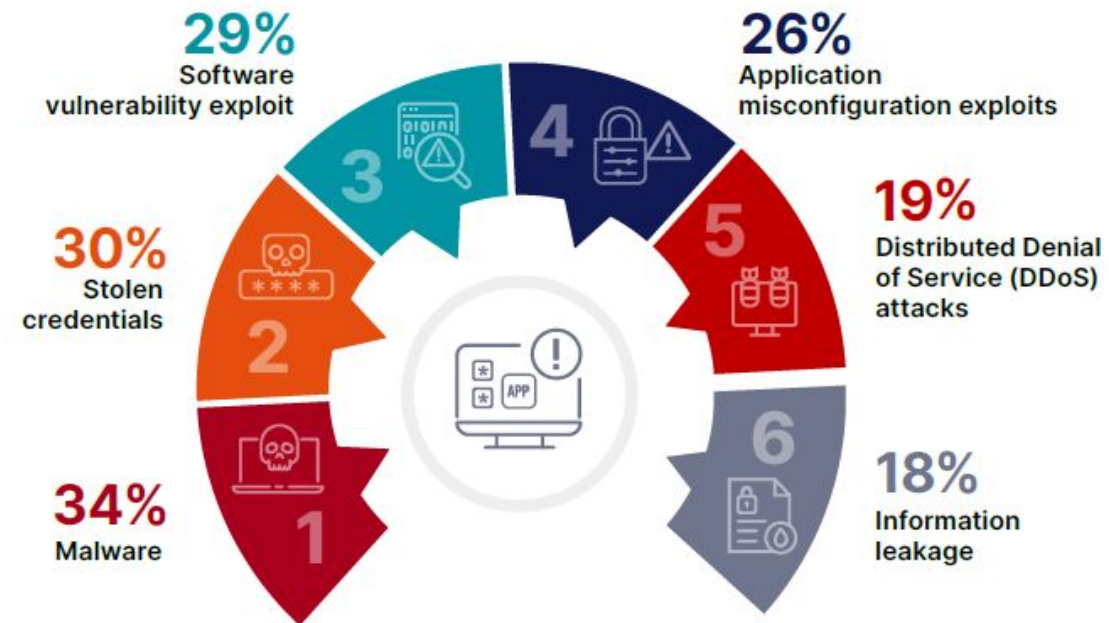
SQL Injection



Malwares

► Which of the following security attacks against applications has your organization experienced over the past 12 months?

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**Web & N/W
Firewall**



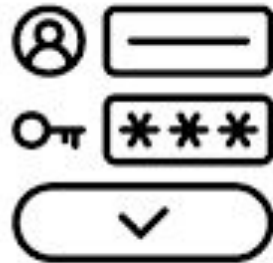
**Keep your S/W
Up to date**



**Backup
your data**



**Keep your
website clean**



**Strong
Password Policy**



**Password
Cracking Tools**



**Scan your website
for vulnerabilities**



**Use
Antivirus**

7. Emerging Trends

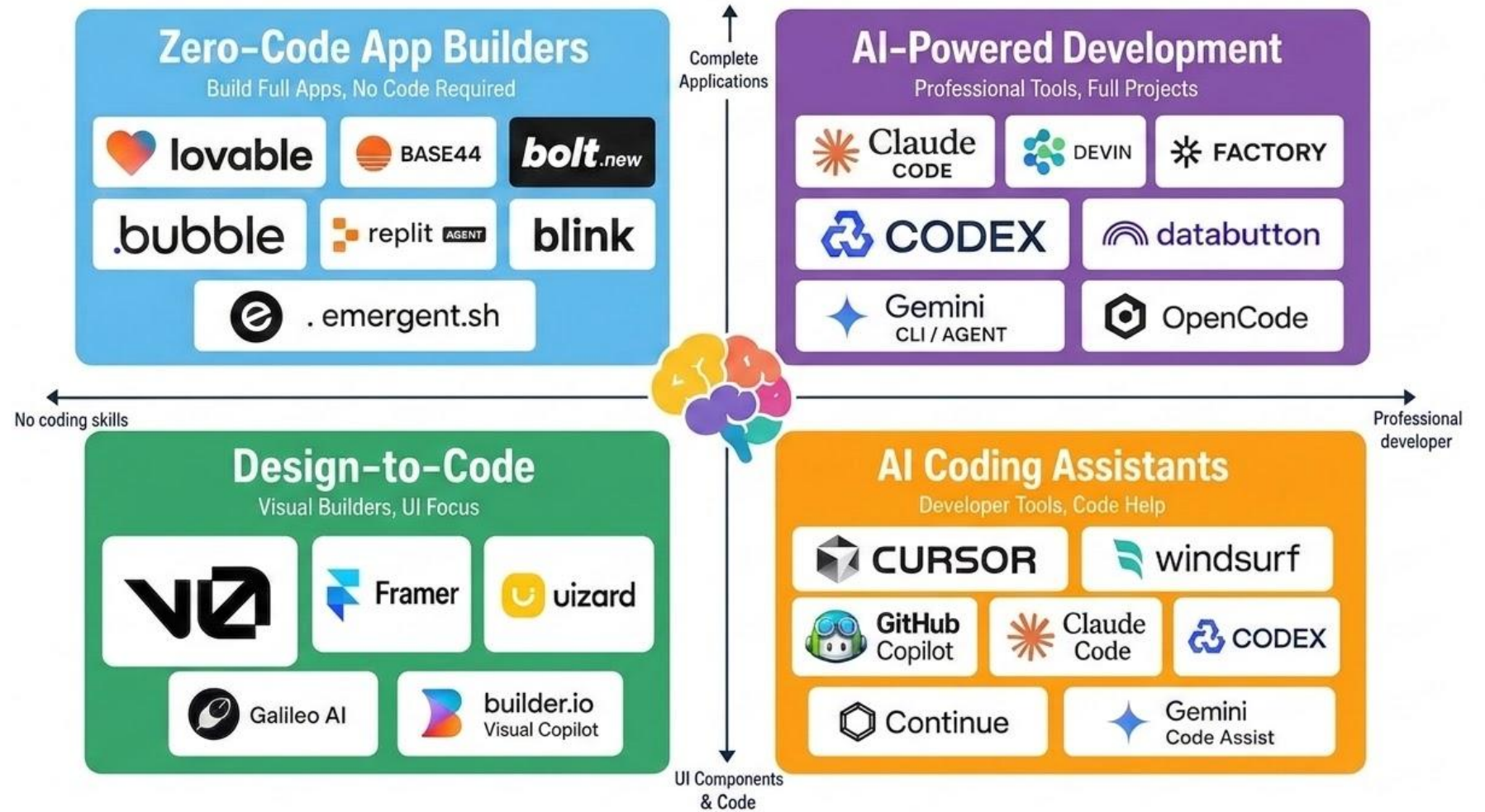
1. ***AI-Powered Web Development*** (OpenAI API, RAG, MCP, LangChain, LangGraph, AI Agents, Prompt Engineering)
2. ***Modern Web Architectures*** (AI-Native Applications, Edge Computing, Serverless Computing, Local-First Applications, WebAssembly)
3. ***Next-Generation User Experiences*** (Agentic UX, Generative UI, Full-Stack Type Safety)
4. ***AI-Assisted Software Engineering*** (GitHub Copilot, Cursor, Claude Code, Vibe Coding, AI-Driven Testing, AI-Driven Code Review)

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"An emerging programming paradigm where developers primarily write code by interacting with code-generating large language models rather than writing code directly."

The Vibe Coding Landscape 2026

The Vibe Coding Revolution: From Idea to Production Software in Natural Language



AustinArmstrong.ai

20+ Major Platforms | From MVP to Enterprise | May 2026



**THANK
YOU**

FOR LEARNING



QNA





next

HTML , CSS

